DiaTwin

Research Concept Brief | TRL 4–5 | dia-twin.vercel.app

Personalised Insulin Dosing Through Circadian-Aware Reinforcement Learning

THE PROBLEM

Over 8 million people with Type 1 Diabetes world-wide rely on insulin pens. Despite widespread CGM adoption, no affordable decision-support tool simultaneously accounts for insulin pharmacodynamics, meal absorption, physical activity, and the time-of-day variation in insulin sensitivity (circadian variability). Fixed correction ratios prescribed by clinicians cannot adapt, leading to chronic dosing errors, elevated hypoglycaemia risk, and high cognitive burden for patients and care teams.

HOW IT WORKS , THE CONTEXTUAL DOSING PIPELINE

Every insulin recommendation integrates all available patient context in a single on-device calculation:

Meal photo → carbs	Camera-based food recognition extracts macronutrients (carbs, protein, fat) and assigns an absorption profile (very fast → very slow) automatically
PK/PD modelling	IOB (Insulin-on-Board) and MOB (Meal-on-Board) computed via gamma-distribution curves; cumulative daily effects and predictive peak-overlap included
Activity adjustment	Logged and wearable-derived activity modifies insulin sensitivity in real time , both acute uptake (GLUT4-independent) and prolonged post-exercise effect
BG + circadian baseline	Current glucose (CGM or finger-prick) corrected against a 24-hour circadian anchor: overnight nadir, dawn phenomenon, and daytime plateau
Dose + insulin type	Suggested units (rounded to 0.1u, 0.5u safety floor) plus the appropriate insulin analogue (rapid vs standard) matched to the meal's absorption profile
Clinician-guided personalisation	All meal–insulin–activity–outcome tuples are logged and displayed in patient charts accessible to supervising clinicians. Doctors review glucose trend graphs and can propose ISF/ICR factor updates directly within the platform. Full ML-driven automated parameter refinement per 2-hour circadian block is planned for the next development phase.

WHAT SETS IT APART

Photo-to-dose	Point camera at a meal → carb estimate → insulin suggestion. No manual entry required
Full context window	Saved meals, doses, activity logs, and BG readings all contribute to every new recommendation — not just the current meal
Circadian adaptation	Time-of-day ISF/ICR personalisation via RL absent from all current pen-user tools
No pump required	Operates entirely with standard insulin pens and finger-prick or CGM data
Equity by design	Designed for settings with limited CGM access , a planned LMIC arm

CURRENT STATUS & CREDENTIALS

Prototype	Live web app + Android app — deployable, demo-ready (dia-twin.vercel.app)
Academic	M.Sc. Digital Health, DIT Germany — grade 1.6 (Excellent, 2025)
Clinical advisory	Prof. Dr. Klaus Parhofer, LMU Munich (Medizinische Klinik IV) — thesis supervisor;
Training data	Engine pre-trained on 50,000+ real meal outcome clinical data pairs; out-of-sample stress-tested against the AZT1D dataset (25 T1D patients, CGM + bolus + carbohydrate data, spanning both pen-based MDI and automated insulin delivery therapy periods)
Researcher	Pharmacist (§11 BApO licence), M.Sc. Digital Health, senior clinical informatics (HL7-FHIR R4)

PROPOSED COLLABORATION

I am seeking a PhD supervision context to design and execute a prospective Pilot RCT (N = 30, 6 months) validating circadian ML adaptation against standard-of-care pen dosing. I welcome discussion around supervision, clinical trial infrastructure, and grant co-development.

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